Lab 2.2 - Inspection Reports and Rectification

App: Microsoft 365 Copilot in **Word**

Purpose: Use Microsoft 365 Copilot in Excel and Word to document both branches of the same work order process — an initial Pass, and an initial Fail followed by rectification and reinspection.

Output: Your `Work-Order-Register.xlsx` updated with this lab's inspection sheets, plus `RPT-021.docx` and `RPT-022-v2.docx` with a clearly versioned reinspection addendum.

Work in `Work-Order-Register.xlsx` — the file you created in Lab 2.1. This folder's `Starter.xlsx` supplies this lab's data as ready-made Excel tables; you copy its sheets across. You do not import the CSV. The `.csv` files are a plain-text copy for the optional practice in Appendix A only.

Where this sits in Module 2

Lab	App	What you produce
2.1	Excel	The work order register
2.2	**Word**	**The two inspection reports**
2.3	Outlook	The approval request to the manager
2.4	PowerPoint	The weekly update to the supervisor

The scenario

A maintenance team inspects roadside drainage on the public road network. Each drain section is raised as a work order, a worker attends the site with a graduated measuring rule, and the reading is compared against the specification maximum of **50 mm silt depth**.

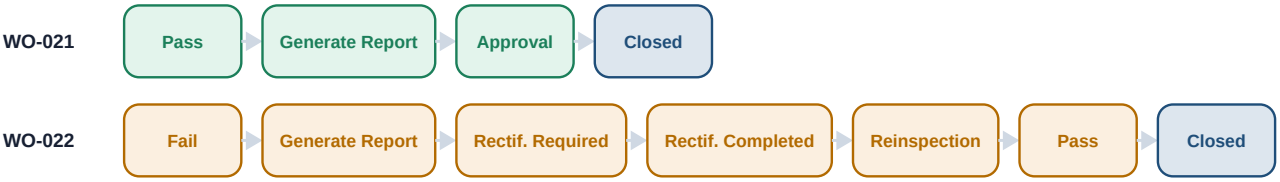
You are continuing the two work orders you opened in Lab 2.1:

Work order	Asset	Location	Initial reading	Outcome
WO-021	DR-17	Jalan Besar, drain section 17	32 mm	**Pass**
WO-022	DR-23	Clementi Avenue 2, drain section 23	74 mm	**Fail**

DR-23 is silted above the limit. A drain silted above the limit cannot carry a storm, which risks flooding the carriageway, so the failure is reported, a desilting is raised as rectification work, and the post is reinspected before the order can be closed. That is the difference this lab documents: one order closes on the first visit, the other needs a second.

Where this lab sits

Both orders continue from Evidence Captured -> Pass / Fail, which you completed in Lab 2.1:



This lab produces the reports at each **Generate Report** box and records the rectification and reinspection states. Closeout and the presentation come in Lab 2.3. You verify each transition yourself; the supplied log is fictional evidence, not proof that a step happened.

Your working file

You carry one register through all three labs.

File	What it is
`Work-Order-Register.xlsx`	**Your own working file**, created in Lab 2.1. You reopen it here and add this lab's sheets to it. Everything you do goes in this file.
`Starter.xlsx`	**This lab's supplied data.** You open it to copy sheets out of it, then close it without saving.

Files in this folder

File	Use
`Lab-Instructions.pdf`	This activity and every copy-ready Copilot prompt.
`Starter.xlsx`	This lab's supplied data, opening on a `Start_Here` sheet, with the `Inspections` and `RectificationLog` tables already built. Copy its sheets into your register.
`mock-data.csv`	The three inspection rows as plain CSV. Not needed — Appendix A only.
`rectification-log.csv`	The two rectification events as plain CSV. Not needed — Appendix A only.

Step 1 - Reopen your register and bring in this lab's data

You need **two** files open: your own register from Lab 2.1, and this lab's supplied starter.

1. Open `Work-Order-Register.xlsx` — the file you saved in Lab 2.1. This is where you work.
2. Open this folder's `Starter.xlsx` as well. It holds the data for this lab:

Sheet	What it holds
`Start_Here`	Where to begin, and what each sheet is for.

Sheet	What it holds
`Inspections`	Three supplied inspection rows as the named table `Inspections`, plus the empty amber columns you will fill.
`Rectification_Log`	The two supplied WO-022 rectification events as `RectificationLog`.
`Branch_Guide`	The two branches above, for reference.
`Check_Your_Work`	The three variances to verify by hand.
`Copilot_Prompt`	The same prompts printed here, ready to copy inside Excel.

3. On `Inspections`, confirm three rows: WO-021 initial, WO-022 initial, and WO-022 reinspection.
4. Copy the `Inspections` and `Rectification_Log` sheets into your register: right-click each sheet tab, choose **Move or Copy**, pick `Work-Order-Register.xlsx` as the destination book, tick **Create a copy**, then **OK**.
5. Check the Lab 2.1 `Work_Orders` and `Status_History` sheets are still there and unchanged. Your register should now hold both labs' sheets.
6. Close `Starter.xlsx` without saving. From here on you work only in `Work-Order-Register.xlsx`.

The data is already a formatted Excel table. You do not need to import anything; ignore the CSVs unless your trainer asks for the import practice in Appendix A. The amber columns `Margin_mm`, `Outcome`, `Evidence_Checked`, `Report_ID`, `Rectification_Status` and `Review_Note` are already in place.

If you do not have the Lab 2.1 register (you missed that lab, or the file is lost), you can work directly in this lab's `Starter.xlsx` and save it as `Work-Order-Register.xlsx`. You will be missing the Lab 2.1 status history, so note that in your review column.

Step 2 - Select the named table

1. Click any cell inside the inspection data.
2. Check the **Name Box**, at the left of the formula bar, shows `Inspections`.

Copilot in Excel works against the table you have selected, so confirm this before prompting.

If the name is wrong or missing, rename it by hand: click inside the table, go to the **Table Design** tab, and type `Inspections` into the **Table Name** field on the far left. It must match exactly, with no space.

Copilot cannot do this for you. Renaming a table is a structural change to the workbook, not a data question, so it sits outside what Copilot in Excel will act on. Asking it to rename the table will not work, and the prompts in this lab refer to `Inspections` by name, so the name has to be right before you prompt.

Step 3 - Fill the inspection table with Copilot

1. With the `Inspections` table still selected, open **Copilot** from the Home tab.
2. Paste the prompt below, or copy it from the `Copilot_Prompt` sheet.
3. Keep the original readings visible. The reinspection is a new row, never an edit to the failed one.

Copy this prompt:

```
Use only the selected Inspections table on the Inspections sheet.

Preserve every Work_Order, Inspection_Round, timestamp, worker, asset, reading,
unit, maximum, evidence ID and source status exactly as supplied.

Add Margin_mm = Silt_Depth_mm - Maximum_mm for each of the three rows and
show the arithmetic. Set Outcome to Pass when Silt_Depth_mm is greater than or
equal to Maximum_mm, otherwise Fail.

Keep all three rows visible, including WO-022's failed first inspection and its
passing reinspection. Do not overwrite the first reading with the second.

Mark missing real-world evidence as NOT SUPPLIED. Do not invent a root cause.
End with DRAFT FOR ENGINEER REVIEW.
```

Step 4 - Verify the arithmetic by hand

Do this before accepting any Copilot output:

Row	Silt depth	Maximum	By hand	Outcome
WO-021 initial	32	50	50 - 32 = **+18**	Pass
WO-022 initial	74	50	50 - 74 = **-24**	Fail
WO-022 reinspection	28	50	50 - 28 = **+22**	Pass

Step 5 - Generate RPT-021 (the pass branch)

1. Open a new Word document, open **Copilot**, and attach your workbook.
2. Paste the prompt below. This drafts the WO-021 report only — WO-022 is a separate run in Step 6.
3. Save as `RPT-021.docx`.

Copy this prompt (run 1 of 3):

```
Using only the attached drainage inspection table and the work order plan
from Lab 2.1, draft an inspection report for WO-021 only.

Include: scope; source records; the factual observation of 32 mm at
DR-17, Jalan Besar; the comparison against the 50 mm criterion; evidence ID
EV-021-1; interpretation and limitations; an action and status table; and
a blank reviewer sign-off.

Report the pass factually. Do not record an actual approval, and do not
claim any real sign-off took place. Mark DRAFT FOR ENGINEER REVIEW.
```

5. Record `Generate Report` as a status row. Record `Approval` **only after a person checks the draft** — the supplied event log contains a mock approval entry, which is training evidence, not a real approval.

Step 6 - Generate RPT-022 (the fail branch)

1. Open a **new** Word document — do not append to RPT-021.

2. Paste the prompt below. This is the initial failure report only; the reinspection is added in Step 7.
3. Save as `RPT-022.docx`, then record `Rectification Required` as a separate status row **after** the report exists.

Copy this prompt (run 2 of 3):

```
Using only the attached drainage inspection table and the work order plan
from Lab 2.1, draft the initial inspection report for WO-022 only.

Include: scope; source records; the factual observation of 74 mm at
DR-23, Clementi Avenue 2; the comparison against the 50 mm criterion, showing the
1.2 mm shortfall; evidence ID EV-022-1; interpretation and limitations;
an action and status table; and a blank reviewer sign-off.

Report the failure factually. Do not infer or assert a root cause: a low
silt depth above the limit is an observation, and the cause is unproven. Do not
mention the reinspection. Mark DRAFT FOR ENGINEER REVIEW.
```

Step 7 - Add the reinspection addendum

1. Read `Rectification_Log`. The fictional record says a desilting was completed.
2. Reopen `RPT-022.docx` and paste the prompt below to append the addendum.
3. Save the result as `RPT-022-v2.docx`. Keep the original failed reading and the first report visible — a pass on reinspection does not erase the initial failure, and does not prove the root cause was found.
4. Record `Rectification Completed`, `Reinspection`, `Pass`, then `Closed` in that order.

Copy this prompt (run 3 of 3):

```
Append a clearly dated reinspection addendum to this WO-022 report, using
only the attached inspection table and rectification log.

The addendum states: the rectification recorded in the log; the
reinspection reading of 28 mm with evidence ID EV-022-2; and the
comparison against the 50 mm criterion, showing it now passes by
0.2 mm.

Keep the original 74 mm failure and EV-022-1 fully visible above the
addendum. Do not edit or delete the first reading. Label the document
RPT-022-v2. State that a pass on reinspection does not by itself prove the
root cause was found. Mark DRAFT FOR ENGINEER REVIEW.
```

Step 8 - Reconcile and save

Reconcile every report ID and evidence ID against Excel. Add a reviewer note wherever evidence images, calibration records or real approvals are not supplied.

Now save your work.

1. In Excel, press **Ctrl+S** (**Cmd+S** on Mac) to save `Work-Order-Register.xlsx`. Confirm the title bar shows that name, and that it now holds the Lab 2.1 sheets (`Work_Orders`, `Status_History`) **and** this lab's (`Inspections`, `Rectification_Log`).
2. Save `RPT-021.docx` and `RPT-022-v2.docx` in the same folder.

3. Keep all three. **Lab 2.3 reopens** `Work-Order-Register.xlsx` and attaches both reports to build the closeout deck.

Copy this prompt:

```
Compare the two reports with the Excel status history. Identify missing,
out-of-order or unsupported transitions. Require Generate Report after the
initial decision. Require approval before WO-021 closure, and rectification
plus reinspection before WO-022 closure.
```

Pass conditions

- Three inspection rows carry the correct arithmetic and outcomes.
- Excel records the ordered states for both branches and keeps the earlier states.
- The pass branch shows report generation and approval before closeout.
- The fail branch shows a failure report before rectification, then a new reading before Pass and Closed.
- The Word reports trace every claim to a supplied record and leave real sign-offs blank.
- No AI output is presented as a real engineering approval.

Appendix A - Optional: import the CSVs yourself

You do not need this if you are using `Starter.xlsx`. Use it only for the import practice, or if the starter file is unavailable. Do not double-click the CSV — that gives you a plain CSV file with no table.

1. Open your `Work-Order-Register.xlsx` from Lab 2.1.
2. **Data** tab -> **Get Data** -> **From File** -> **From Text/CSV**.
3. Select `mock-data.csv` and choose **Import**.
4. Check **Delimiter** is **Comma** and the columns split correctly, then **Load To...** -> **Table** -> **New worksheet** -> **OK**.
5. Click inside the new table. On **Table Design**, set **Table Name** to `Inspections`. Rename the sheet tab to `Inspections`.
6. Repeat steps 2-5 for `rectification-log.csv`, naming that table `RectificationLog` on a sheet named `Rectification_Log`.
7. Add the columns `Margin_mm`, `Outcome`, `Evidence_Checked`, `Report_ID`, `Rectification_Status` and `Review_Note` to the right of the `Inspections` table.
8. Confirm `Silt_Depth_mm` and `Maximum_mm` are right-aligned, so Excel read them as numbers. If not, select the column and use **Data -> Text to Columns -> Finish**.
9. Save as **Excel Workbook (*.xlsx)**, keeping the Lab 2.1 sheets.

All records in this lab are fictional training data. They describe no actual facility, person or organisation. Values are teaching assumptions, not approved design criteria. Do not use any output to operate, maintain or approve real equipment.